James Akl

Scientist & Engineer

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Worcester, MA, USA

SUMMARY

I bring a strong background in **Engineering** and **Mathematics**, with expertise in **Robotics, AI**, and **Autonomous Systems**.
I hold 5 years of experience in **R&D, rapid prototyping, technology demonstration**, and **commercialization** with emphasis on:

- **analyzing problems** and requirements, as well as devising **creative solutions** to address them.
- **software development, mathematical modeling**, autonomous **system design**, and **algorithm design**.
- **project leadership**, planning, and management in technical and R&D settings.

My work is **structured, meticulous**, and **rigorous** — I am routinely invested in raising and maintaining **high standards**.
I seek opportunities in **technology, research**, and **industry** to create value via autonomous and intelligent systems.

EDUCATION

PhD Robotics Engineering — Worcester Polytechnic Institute — **2019–2023**
Focus: Autonomous robots and systems · Dissertation: "Robot Autonomy for Scrap Cutting in Metal Recycling"

MS Robotics Engineering — Worcester Polytechnic Institute — **2019–2022**
Focus: Robot dynamics and control, Artificial intelligence, Machine learning, Applied mathematics, Data science

Bach Mechanical Engineering — Lebanese American University — **2015–2019**
Focus: Robotics, Software, Control, Mechanics, Sensors · Thesis: "Robotic Manipulator for VR Haptic Feedback"

Minor Mathematics — Lebanese American University — **2015–2018**
Focus: Analysis, Optimization, Linear algebra, Numerical analysis, Probability, Statistics, Modeling, Algorithms

EXPERIENCE

Postdoctoral Scientist · **Amazon** · Innovation Lab (Fulfillment Technologies & Robotics)

2024/03 – PRESENT: Conducting **robotics/AI research** incorporating **robot learning, perception, control, manipulation, grasping, human–robot interfacing**. Leading a prototyping team and developing innovative **robotic fulfillment systems**.

Applied Scientist · **Amazon** · Innovation Lab (Fulfillment Technologies & Robotics)

2023/08 – 2024/03: Conducted **robotics R&D** to design/implement algorithms for autonomous fulfillment and logistics. Developed **robotic manipulation** capabilities using **3D perception, deep learning**, task and motion **planning**, and **control**.

Robotics Researcher · **Worcester Polytechnic Institute** · Manipulation & Environmental Robotics Lab

2019/08 – 2023/08: Solved challenging technical problems by developing **innovative solutions** and **novel algorithms**.

Managed and lead **research projects**, supervised MS students, and collaborated in multidisciplinary teams.
Developed **software** for prototypes, simulations, and experiments (C++, Python, ROS, Bash, Linux, Git, PCL, OpenCV, …).
Engineered **autonomous systems** for industrial applications using **robot vision, control, planning**, and **machine learning**.
My work resulted in **8 research publications** in top-tier conferences/journals, as well as **3 patents** filings.

Research Mentor · **Worcester Polytechnic Institute** · Robotics Engineering Department / NSF REM

2022/05 – 2022/07: Mentored students in **research methodology, data collection/processing**, image processing, and **hands-on experiments**. Presented to the NSF REM Program my strategies for **effective mentoring** of research trainees.

Teaching Assistant · **Worcester Polytechnic Institute** · Robotics Engineering Department

2019/08 –2020/05: Assisted in **delivery, labs**, and **grading** of robotics courses ('Actuation', 'Sensing', 'Intro. to Robotics').

Visiting Researcher · **University of Ottawa** · Mathematical Modeling & Physical Experiments

2018/06 – 2018/08: **Modeled** the vibrations of ribbed plates, and **experimentally** evaluated **analytical models** against **numerical models**. My work resulted in **2 research publications** in reputable journals in the field of mechanical vibrations.

Research Assistant · **Lebanese American University** · Multiphysics Modeling & Numerical Simulation

2017/11 – 2019/03: **Demonstrated** the functionality of the 'ionic buoyancy engine' for underwater propulsion using a multiphysics **numerical model**. My work resulted in a publication at a top-tier conference in the field of smart materials.

SKILLS

Software	C++ · Python · Bash · Git · Linux · CMake · ROS2 · OpenCV · PCL · Open3D · Eigen · OctoMap · PyTorch
	SciPy · scikit-learn · scikit-image · matplotlib · Isaac Sim · Gazebo · MATLAB/Simulink · Mathematica
3D Tools	Blender · MeshLab · Nvidia Omniverse · Meshmixer · Meshroom · F3D · Fusion 360 · SolidWorks
Robotics	Kinematics · Dynamics · Control · Motion planning · Calibration · Simulation · Manipulation & Grasping
	Bin picking / packing · Logistics / fulfillment · Metal cutting · Waste sorting · Soft robot control
AI & ML	Deep learning · Perception · Computer vision · Synthetic data · Search algorithms · Probabilistic reasoning
Platforms	Universal Robots · Franka Emika · Nvidia Jetson · Intel RealSense · Arduino · Raspberry Pi
Research	Surveys/reviews · Mathematical modeling · Simulations · Physical experiments · Field work
Publishing	LaTeX · Technical writing · Photo-editing · Illustration/diagrams · Video-editing · MS Office
Social	Presentations · Communication (technical, general) · Collaboration (cross-disciplinary)
Language	English (native fluency) · French (professional fluency) · Arabic (professional fluency)

PUBLICATIONS

[2024][1] **Robot Perception for Transporting Items from Bins with Identical Objects to Bins with Mixed Objects** in ***Amazon CSS***
[2023][1] **Cut Sequencing Algorithm for Safely Disassembling Large Structures** in ***IEEE CDC***
[2023][1] **Vision-based Oxy-fuel Torch Control for Robotic Metal Cutting** in ***IEEE IROS***
[2023][1] **CNN-based Task State Estimation for Safer Automation of Oxy-fuel Metal Cutting** in ***IEEE CASE***
[2023][1] **Feature-driven Next View Planning for Cutting Path Generation in Robotic Metal Scrap Recycling** in ***IEEE T-ASE***
[2022][7] **VisDA 2022 Challenge: Domain Adaptation for Industrial Waste Sorting** in ***PMLR/NeurIPS Competition Track***
[2022][4] **ZeroWaste Dataset: Towards Deformable Object Segmentation in Cluttered Scenes** in ***IEEE CVPR***
[2021][1] **Towards Robotic Metal Scrap Cutting: A Novel Workflow and Pipeline for Cutting Path Generation** in ***IEEE CASE***
[2021][2] **ECNNs: Ensemble Learning Methods for Improving Planar Grasp Quality Estimation** in ***IEEE ICRA***
[2020][1] **Comparing & Optimizing Analytical, Numerical & Experimental Vibration Models for […]** in ***IStructE/Elsevier Structures***
[2019][4] **Simplified Setup for the Vibration Study of Plates with Simply-supported Boundary Conditions** in ***Elsevier MethodsX***
[2019][1] **Ionic Buoyancy Engines: Finite Element Modeling & Experimental Validation** in ***SPIE EAPAD*** (*Smart Structures*)

[1–4] **Superscripts indicate authorship position in co-authored publications.**

PATENTS

Autonomous Robotic Cutting System · 18/377,196 · Filed Oct 5, 2023
Feature-driven Next View Planning of 3-Dimensional Surfaces · 18/119,547 · Filed Mar 9, 2023
Salvage Metal Cutting Robot · 17/721,553 · Filed Apr 15, 2022

VOLUNTEERING

Scholarly Reviewer · **IEEE** and **Sage Publications** · International Journals & Conferences

2021/06 – PRESENT: Conducting scholarly peer reviews for reputed journals (IJRR, RA-L, T-ASE, TMECH) and conferences (ICRA, IROS, CDC, CASE) in the areas of robotics, AI/ML, control, automation, mechatronics, and autonomous systems.

Faculty Candidate Reviewer · **Worcester Polytechnic Institute** · Robotics Engineering Department

2023/02 – 2023/03: Attended **weekly seminars** of faculty candidates to the WPI Robotics Engineering Department. Submitted **written evaluations** of candidates' potential in research, teaching, supervision, funding, and inclusive work.